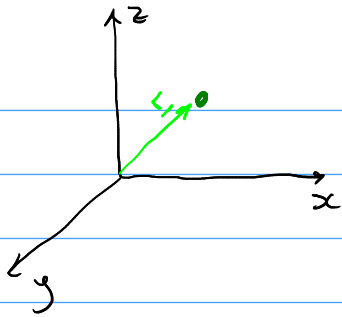




$$f = r^n = \|x\|^n \cdot x$$

$$r = \|x\|$$

$$\|x\|^2 (x + z)$$

$$\nabla \phi + z$$

$$f = x^2$$

$$r = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \Rightarrow r(x_1, x_2, x_3)$$

$$\|r\| = \sqrt{x_1^2 + x_2^2 + x_3^2}$$

$$r \cdot r = \|r\| \|r\| \cdot \cos 0 = \|r\|^2 = r^2$$

$$r^2 = x_i \cdot x_i$$

$$r = \sqrt{x_1^2 + x_2^2 + x_3^2}$$

$$\frac{\partial r}{\partial x_i} \cdot \frac{\partial r^2}{\partial x_i} = \frac{\partial r(x_i)^2}{\partial x_i} = 2r \cdot \frac{\partial r}{\partial x_i}$$

cambiamos a j el subindice porque i ya era dami

$$\frac{\partial (x_j x_j)}{\partial x_i} = \frac{\partial x_j}{\partial x_i} x_j + x_j \frac{\partial x_j}{\partial x_i}$$

$$\delta_{ji} x_j + x_j \delta_{ji} = x_i + x_i = 2x_i$$

$$= x_i + x_i = 2x_i$$

$$2r \frac{\partial r}{\partial x_i} = 2x_i \Rightarrow \frac{\partial r}{\partial x_i} = \frac{x_i}{r}$$

$$r^2 = x_i x_i \Rightarrow r = \sqrt{x_i x_i}$$

$$\frac{\partial r}{\partial x_i} = \frac{1}{2} \frac{\frac{\partial}{\partial x_i} (x_j x_j)}{\sqrt{x_k x_k}} = \frac{1}{2} \frac{2x_i}{\sqrt{x_k x_k}} = \frac{x_i}{r}$$

$$u_i v_i = u_j v_j = u_g v_g \quad \text{Bij}_{jkem} = A_{ij} C_{jkem} \rightarrow \text{orden 4}$$

$$\delta_{ij} v_j = v_i \quad \text{Bij}_{jkem} = A_{ij} C_{jkem} \rightarrow \text{orden 6}$$

$$e_{ijk} = \delta$$

3D

orden 1 escalar x vector

$$\delta_{ij} \frac{\partial x_i}{\partial x_j} \rightarrow \frac{\partial x_1}{\partial x_1}, \frac{\partial x_1}{\partial x_2}, \dots$$

escalar

$$a) \nabla \cdot (r^n \underline{r}) = \frac{\partial}{\partial x_i} (r^n \underline{r})_i = \frac{\partial}{\partial x_i} (r^n \tilde{r}_i) = \frac{\partial r^n}{\partial x_i} x_i + r^n \frac{\partial x_i}{\partial x_i}$$

$$= n \cdot r^{n-1} \cdot \frac{\partial r}{\partial x_i} x_i + r^n \cdot 3 = n \cdot r^{n-1} \cdot \frac{x_i \cdot x_i}{r} + 3 r^n =$$

$$= n \cdot r^n + 3 r^n = r^n (n+3) \quad \left(\underline{A} \underline{x} \right)_k = A_{kp} x_p$$

$$b) \left[\nabla \times (r^n \underline{r}) \right]_i = \epsilon_{ijk} \frac{\partial (r^n \underline{r})_k}{\partial x_j} = \epsilon_{ijk} \left[\frac{\partial r^n}{\partial x_j} x_k + r^n \frac{\partial x_k}{\partial x_j} \right] =$$

$$= \epsilon_{ijk} n \cdot r^{n-1} \cdot \frac{\partial r}{\partial x_j} x_k + \epsilon_{ijk} r^n \delta_{kj}$$

$$= n \cdot r^{n-2} \underbrace{\epsilon_{ijk} x_j \cdot x_k}_0 + \underbrace{\epsilon_{ijj} r^n}_0 = [0]_i$$

$$\underline{\nabla} \times (r^n \underline{r}) = \underline{0}$$